Martin Long Hoang Tran

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Software Engineer Intern

EDUCATION

Cal State University, Fullerton - Bachelor of Science, Computer Science **Expected in Dec 2027**

EXPERIENCE

Software Development Intern, OpenQQuantify - Vienna, VA May 2025 - Oct 2025

- Boosted FAQ accuracy by **50%** and automated query triage by building an AI-powered health assistant.
- Built a Python ETL pipeline processing **50K+ drug label records**, reducing manual data preparation by **15 hrs/week**.
- Engineered **3 Flask REST endpoints** for 3-stage query resolution (structured lookup, **MiniLM vector retrieval**, **FLAN-T5** fallback), cutting latency by **>99%** (**1.24s to <9ms**) and enforcing **19 medical safety checks**.

PROJECTS

RevenueTwin | AI Revenue Audit Platform | LPL Financial Hackathon (1st Place Track Winner)

- Built **Bedrock Converse** agent (**Nova Lite**, **4-turn** loop) with read-only tool access isolating fee math.
- Audited **50 households** (**132 accounts**) in **<2 ms**, catching **8/8 seeded fee anomalies** and **~$45K+/yr** billing gaps.
- Enforced **26 engine validation rules**, **4 prompt guardrails**, and **7 JSON Schema** contracts across fee tiers.
- Validated deterministic Python fee engine and Bedrock tool dispatch via **27 unit tests** and CLI harness.
- Github: https://github.com/martintran23/RevenueTwin

WeCalendar | Shared Calendar & Notes Platform | Full-Stack Web Application

- Designed **PostgreSQL** schema with **53 RLS policies** across **14 tables**, workspace RPCs, and **20 migrations**.
- Engineered **Supabase real-time** sync across **11 tables** (9 live channels) for calendar, notes, and places.
- Built a shared **Leaflet** map with GPS filtering and **500 ms Esri/Nominatim** search, cutting lookup latency from **8s**.
- Implemented **TipTap** rich-text notes with shared/private RLS, event linking, and **183 Vitest tests** (**~80% coverage**).
- Live: https://we-calendar.vercel.app/

WattTime | Solar Energy Optimization Platform | CloudHacks 2026 @ UCI

- Built a full-stack application that optimizes appliance usage using solar forecasts and location-based data.
- Designed an event-driven pipeline with **AWS API Gateway** and **Step Functions** for scalable optimization workflows.
- Reduced response time by **66% (45s → 15s)** by refining LLM prompts and improving inference efficiency.
- Implemented geocoding with **Amazon Location Service** and deployed via **AWS Amplify** with CI/CD.
- GitHub: https://github.com/martintran23/zotwatt

PROFESSIONAL DEVELOPMENT

System Engineer Participant, NASA L'SPACE Program - Tempe, AZ May 2025 - Aug 2025

- Researched technologies aligned with NASA's Technology Taxonomy, including radiation modeling, mitigation, and life support systems for long-duration missions.
- Analyzed systems engineering trade-offs for spacecraft safety, survivability, and mission-critical operations.
- Evaluated technical proposals using NASA review criteria and provided structured feedback.
- Designed and refined CAD models using **Siemens NX** to support system-level concepts.

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL
Frameworks & Tools: Next.js, React, Node.js, FastAPI, Flask, Tailwind, Vitest, Postman, Git, Streamlit, boto3
Databases: MongoDB, PostgreSQL, Supabase
Cloud & DevOps: AWS (Bedrock, S3, API Gateway, Step Functions, Amplify), Vercel, REST APIs, CI/CD, Docker, Linux/Unix